

Polynomial regression

Machine Learning Assignment 1

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github.com/TheOrangeCarXX/polynomial_regression

1. Data, model, and evaluation procedure

The assignment asks for two polynomial regression models with different input counts and degree limits. Each personalized training file contains 1,000 labeled rows. Each corresponding test file contains 1,000 rows without target values. The four input files were checked for the required column order, numeric values, and missing or infinite values before fitting.

Dataset	Input columns	Degrees evaluated	Train / test rows
Var1	x1 through x6	1 through 10	1,000 / 1,000
Var2	x1 through x3	1 through 20	1,000 / 1,000

Polynomial expansion and Ridge regression

A total-degree polynomial of degree d contains every product of the inputs whose exponents sum to at most d . With three inputs and $d = 2$, the nonconstant terms are x_1 , x_2 , x_3 , x_1^2 , x_1x_2 , x_1x_3 , x_2^2 , x_2x_3 , and x_3^2 . A separate intercept supplies the constant term. The number of nonconstant terms for p inputs is $C(p + d, d) - 1$, where C denotes the binomial coefficient.

`PolynomialFeatures` builds these terms. `StandardScaler` subtracts each term's training mean and divides by its training standard deviation. Ridge estimates the coefficients by minimizing the sum of squared prediction errors plus alpha times the sum of squared non-intercept coefficients. Alpha controls the strength of this penalty. It limits large coefficients when polynomial columns are correlated. The intercept is unpenalized.

Selection, holdout, and refitting

A shuffled split with seed 42 reserves 200 rows as a holdout and uses 800 rows for development. Five-fold cross-validation with seed 43 divides the development rows into five groups. Each fold fits on 640 rows and validates on the remaining 160. Every fold fits its own polynomial expansion and scaler. Validation and holdout rows never contribute to those fitted preprocessing statistics.

The search evaluates 50 logarithmically spaced alpha values from 0.0001 to 10,000 for every allowed degree. One singular value decomposition per degree and fold reuses the Ridge solution across alphas. The degree-and-alpha pair with the lowest mean validation MSE is selected. That pair fits the 800 development rows and is assessed once on the untouched holdout. Finally, the same pair fits all 1,000 labeled rows and predicts the supplied test rows.

MSE is the mean squared difference between actual and predicted targets, so smaller values indicate smaller errors. R^2 compares squared prediction error with the target's variation about its mean. A value of 1 is a perfect fit; a value below 0 is worse than predicting that evaluation set's mean. CV R^2 is the mean of the five fold scores.

2. Var1: degree and regularization selection

The search evaluated every total degree from 1 to 10 and all 50 alpha values on the same five development folds. For each degree, the table shows the alpha with the lowest mean validation MSE. The final selection minimizes this MSE over both degree and alpha.

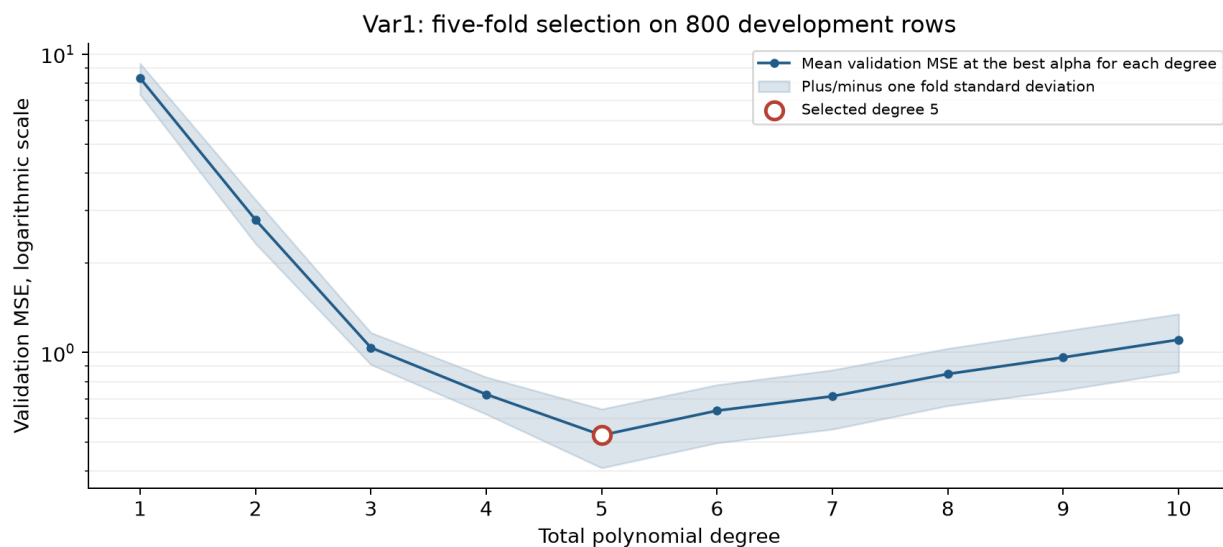


Figure 1. Validation MSE on a logarithmic axis. The shaded band is plus/minus one standard deviation across the five folds, not a confidence interval.

Degree	Alpha	Mean CV MSE	Fold SD	Mean CV R2
1	51.7947	8.308984	0.994499	0.093521
2	11.514	2.786435	0.468612	0.696228
3	5.42868	1.038719	0.127192	0.886137
4	11.514	0.724363	0.103153	0.919839
5	24.4205	0.528587	0.117995	0.941952
6	51.7947	0.638766	0.140793	0.929634
7	109.854	0.713834	0.160483	0.921486
8	232.995	0.847767	0.184306	0.906211
9	339.322	0.963495	0.216113	0.893691
10	494.171	1.104303	0.242722	0.877910

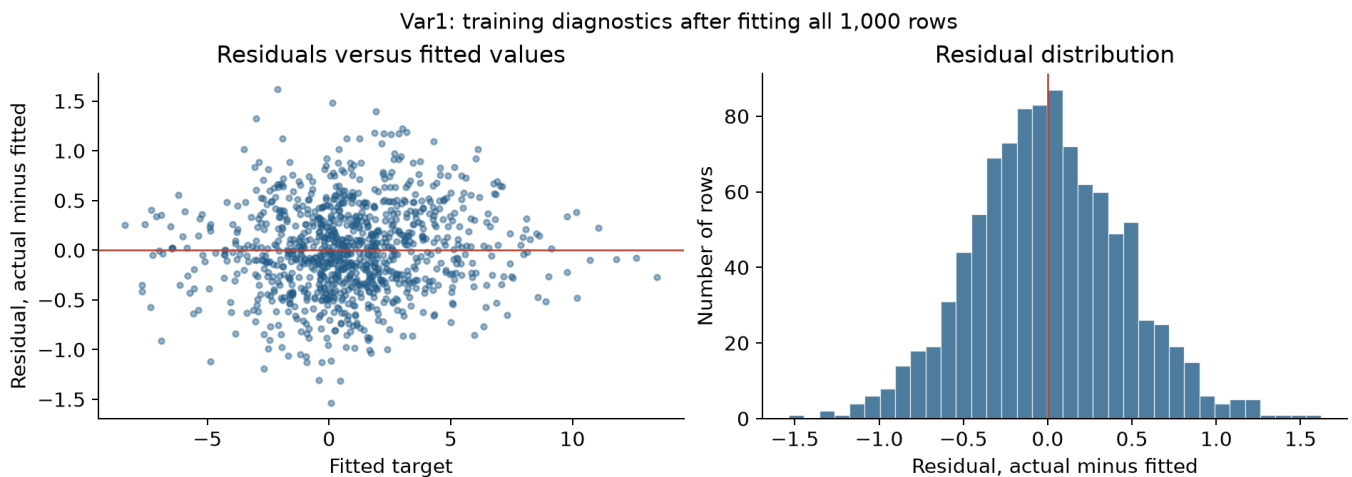
The lowest evaluated mean CV MSE is 0.528587, at degree 5 and alpha 24.420531. This identifies the best pair among the evaluated configurations. The CV score also guided selection, so the separate holdout score is used for the final assessment.

3. Var1: assessment and final model

The selected degree 5 generates 461 nonconstant polynomial terms. Ridge adds a separate intercept, giving 462 fitted coefficients in total. The alpha is 24.42053095. The model fitted on 800 development rows was evaluated once on the 200 holdout rows. Degree and alpha remained fixed afterward.

Evaluation	Rows per evaluation	MSE	R2
Selected pair, mean CV	160 per fold	0.528587	0.941952
Development training	800	0.193723	0.978955
Untouched holdout	200	0.445705	0.962392
Final model training	1,000	0.203276	0.979125

The final model then fits all 1,000 labeled rows with the selected settings. Its training score measures how closely it fits those rows. The holdout result above belongs to the earlier 800-row model; it is not a test score for the refitted model.



The residual plot checks for systematic training errors and unusually large errors. The fitted intercept makes the mean training residual approximately zero. A roughly centered histogram alone does not establish performance on unseen rows. The independent holdout result provides the direct evidence available for that purpose.

Test inference

The saved final pipeline transforms the 6 inputs in each test row using its training-fitted polynomial expansion and scaler, then predicts y . The output `BT2024103_pred_var1.csv` contains one y column and 1,000 rows in the original test-file order, with no index column. Hidden test targets are unavailable, so this report does not claim test MSE or test R2.

4. Var2: degree and regularization selection

The search evaluated every total degree from 1 to 20 and all 50 alpha values on the same five development folds. For each degree, the table shows the alpha with the lowest mean validation MSE. The final selection minimizes this MSE over both degree and alpha.

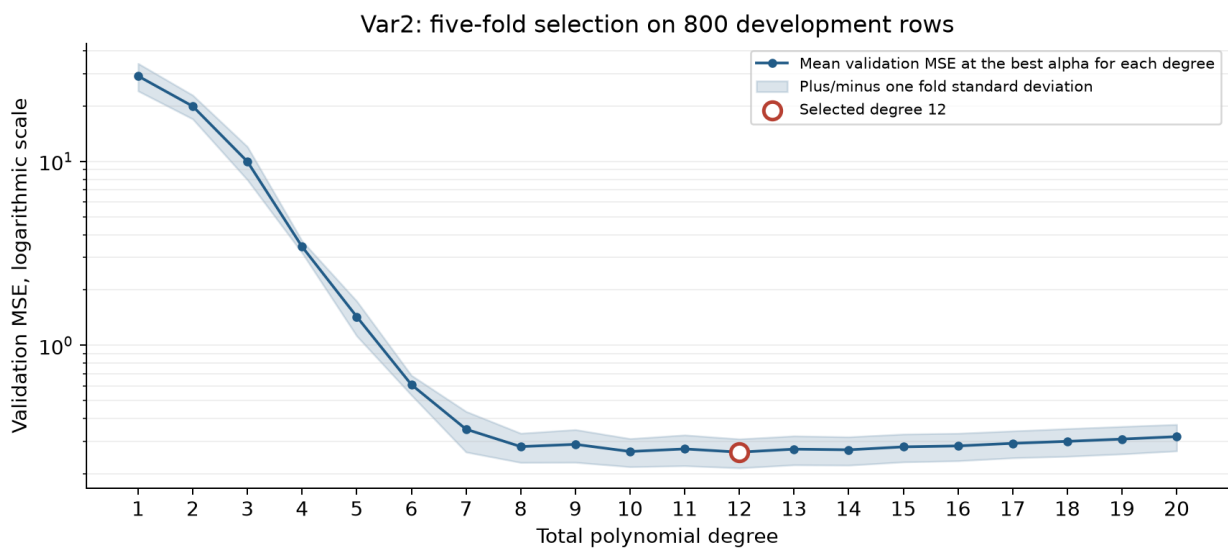


Figure 2. Validation MSE on a logarithmic axis. The shaded band is plus/minus one standard deviation across the five folds, not a confidence interval.

Degree	Alpha	Mean CV MSE		Fold SD	Mean CV R2	
1		16.7683	29.084166		4.954669	0.256644
2		3.72759	19.929319		2.944276	0.488921
3		0.568987	9.972992		2.040794	0.744494
4		0.0001	3.439143		0.255385	0.910998
5		0.126486	1.432758		0.307874	0.963236
6		0.0013895	0.611912		0.075162	0.984142
7		0.0001	0.350286		0.087111	0.991030
8		0.0596362	0.281927		0.050921	0.992704
9		0.26827	0.289924		0.058520	0.992528
10		0.828643	0.265154		0.046162	0.993149
11		1.20679	0.273715		0.052091	0.992942
12		1.75751	0.262967		0.047133	0.993214
13		2.55955	0.273047		0.048929	0.992954
14		3.72759	0.270800		0.047537	0.993015
15		3.72759	0.280955		0.048494	0.992748
16		5.42868	0.284311		0.048423	0.992667
17		5.42868	0.293666		0.048963	0.992421
18		5.42868	0.301023		0.051633	0.992236
19		5.42868	0.309668		0.052732	0.992012
20		7.90604	0.319286		0.052463	0.991766

The lowest evaluated mean CV MSE is 0.262967, at degree 12 and alpha 1.7575106. This identifies the best pair among the evaluated configurations. The CV score also guided selection, so the separate holdout score is used for the final assessment.

5. Var2: assessment and final model

The selected degree 12 generates 454 nonconstant polynomial terms. Ridge adds a separate intercept, giving 455 fitted coefficients in total. The alpha is 1.757510625. The model fitted on 800 development rows was evaluated once on the 200 holdout rows. Degree and alpha remained fixed afterward.

Evaluation	Rows per evaluation	MSE	R2
Selected pair, mean CV	160 per fold	0.262967	0.993214
Development training	800	0.157365	0.995993
Untouched holdout	200	0.254243	0.994152
Final model training	1,000	0.160039	0.996019

The final model then fits all 1,000 labeled rows with the selected settings. Its training score measures how closely it fits those rows. The holdout result above belongs to the earlier 800-row model; it is not a test score for the refitted model.

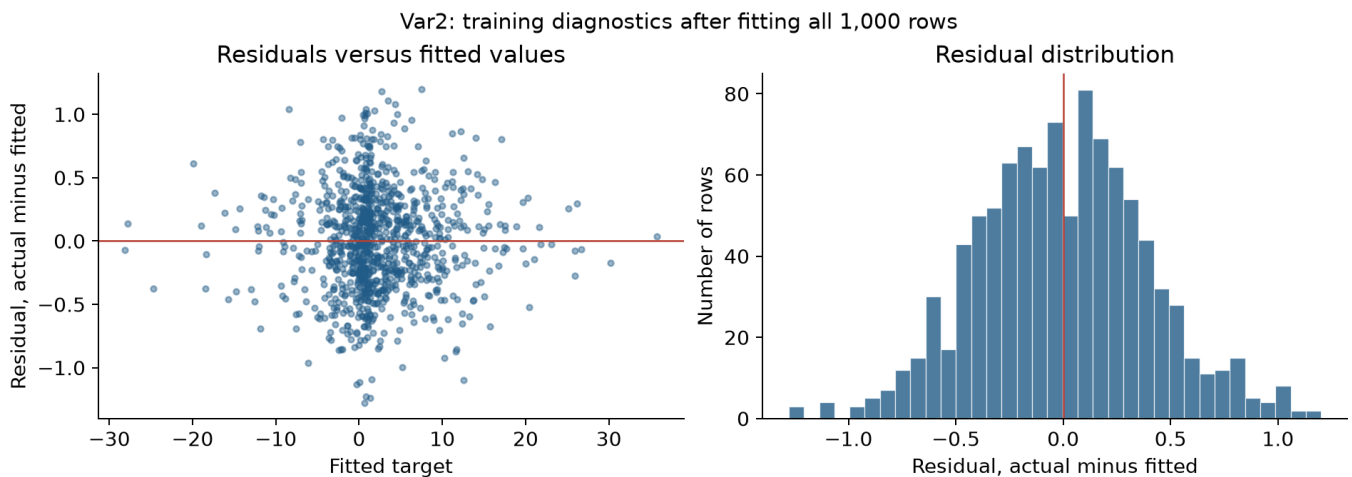


Figure 4. Training residuals after the final 1,000-row refit. A residual is the observed target minus the fitted target.

The residual plot checks for systematic training errors and unusually large errors. The fitted intercept makes the mean training residual approximately zero. A roughly centered histogram alone does not establish performance on unseen rows. The independent holdout result provides the direct evidence available for that purpose.

Test inference

The saved final pipeline transforms the 3 inputs in each test row using its training-fitted polynomial expansion and scaler, then predicts y . The output `BT2024103_pred_var2.csv` contains one y column and 1,000 rows in the original test-file order, with no index column. Hidden test targets are unavailable, so this report does not claim test MSE or test R2.

6. Reproduction, deliverables, and interpretation

The repository contains all training and inference code, both prediction CSVs, four generated figures, full search results, final saved models, and this report's Python source. Input datasets are obtained using the download link in the assignment brief and placed together in `data/`. The README gives the exact four expected filenames.

Reproduction commands

```
python -m pip install -r requirements.txt
python ml_assignment_solution.py --data-dir data
python generate_report.py
```

The first script reruns the full search and saves results under `results/`, figures under `figures/`, and final models under `models/`. The report generator reads those saved results and images. Its tables and numerical descriptions therefore come from the same run as the prediction files. Paths are command-line arguments or relative to the script location; no personal Windows path is required.

Inference without repeating model selection

```
python ml_assignment_solution.py --data-dir data --predict-only
```

This command loads the supplied final pipelines and regenerates both prediction files. It requires only the two test input CSVs. Each saved pipeline contains the polynomial transform, fitted scaler, and fitted Ridge model.

Results and limits

Dataset	Degree	Alpha	Holdout MSE	Holdout R2
Var1	5	24.4205	0.445705	0.962392
Var2	12	1.75751	0.254243	0.994152

These are the lowest-CV-MSE configurations among the degrees and alphas evaluated. They are not a claim of a globally optimal polynomial. Selection CV scores can be optimistic because the search chooses their minimum. The holdout avoids that reuse during selection, although a single 200-row split still has sampling uncertainty. The final full-data refit has no separate observed test score because the assignment withholds the test targets.

Recorded environment

python 3.12.14; numpy 2.3.5; pandas 3.0.1; scipy 1.18.1; scikit-learn 1.9.1; matplotlib 3.11.2; joblib 1.6.0. Pinned dependencies appear in `requirements.txt`. Input SHA-256 hashes and the split settings appear in `results/metrics.json`. Small floating-point differences across numerical libraries or machines are possible.

References

Assignment 1 brief supplied with the datasets. scikit-learn documentation for [PolynomialFeatures](#), [StandardScaler](#), [Ridge](#), and [KFold](#).